THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 6 — MATHS · NCERT-TYPE FULL-YEAR PAPER SET 1

Full Syllabus (Chapters 1–10) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. In the Virahanka (Fibonacci) sequence $1, 2, 3, 5, 8, \ldots$ the 7th term is: 1

(a) 29 **(b)** 21 **(c)** 13 **(d)** 34

2. What is the smaller angle between the hands of a clock at 3 o'clock? 1

(a) 270 **(b)** 60 **(c)** 120 **(d)** 90

3. Add 14 and its reverse 41. The sum is: 1

(a) 77 **(b)** 44 **(c)** 55 **(d)** 66

4. In a pictograph one symbol stands for 10 units. How many symbols (counting a half symbol as half) are needed to show 45? 1

(a) 450 **(b)** 4.5 **(c)** 4 **(d)** 5

5. Which of the following is divisible by 4? 1

(a) 3518 **(b)** 3516 **(c)** 3517 **(d)** 3519

6. The area of a triangle with base 10 cm and height 8 cm is (in cm^2): 1

(a) 50 **(b)** 30 **(c)** 40 **(d)** 48

7. $\frac{1}{2} + \frac{1}{3} =$ 1

(a) $\frac{1}{6}$ **(b)** $\frac{5}{6}$ **(c)** 1 **(d)** $\frac{2}{5}$

8. A circle has centre O and radius 5 cm. A point P is 9 cm from O. The point P lies: 1

(a) inside the circle **(b)** outside the circle

(c) on the circle **(d)** at the centre of the circle

9. A rhombus (not a square) has how many lines of symmetry? 1

(a) 0 **(b)** 1 **(c)** 2 **(d)** 4

10. -12 + 7 = 1

(a) -7 **(b)** -3 **(c)** 9 **(d)** -5

11. A complete graph has 3 points, each joined to every other point. How many lines are there? 1

(a) 3 **(b)** 6 **(c)** 4 **(d)** 9

12. A ray turns anticlockwise through 20° from the horizontal line to the right. The type of angle formed is: 1

(a) acute **(b)** straight **(c)** right **(d)** obtuse

13. Kaprekar step: for 3752, (largest arrangement) - (smallest arrangement) of its digits is: 1

(a) 5166 **(b)** 5184 **(c)** 5265 **(d)** 5175

14. A bar graph has scale 1 cm = 5 students. A bar showing 40 students will have height (in cm): 1

(a) 8 **(b)** 200 **(c)** 35 **(d)** 10

15. Which pair is a pair of co-prime numbers? 1

(a) 6 and 9 **(b)** 8 and 15 **(c)** 10 and 25 **(d)** 12 and 18

16. A right-angled triangle has legs 6 cm and 8 cm. Its area (in cm^2) is: 1

(a) 30 **(b)** 18 **(c)** 32 **(d)** 24

17. $\frac{3}{4} - \frac{1}{3}$ = 1

(a) $\frac{1}{3}$ **(b)** $\frac{5}{12}$ **(c)** 2 **(d)** $1\frac{1}{12}$

18. Two points A and B are 8 cm apart. The perpendicular bisector of AB meets AB at a point which is at what distance from A (cm)? 1

(a) 5 **(b)** 8 **(c)** 4 **(d)** 3

19. The smallest angle through which a regular pentagon must be rotated to coincide with itself is (degrees): 1

(a) 82 **(b)** 72 **(c)** 62 **(d)** 102

20. 5 - 12 = 1

(a) -5 **(b)** -7 **(c)** 17 **(d)** -9

SECTION — B (Very Short Answer Type) — 6 × 2 = 12

Answer in about 20–30 words each.

21. Write the first 10 terms of the Virahanka sequence that starts with 1, 1. 2

22. Find $\angle AOC$ if $\angle AOB = 46^\circ$ and $\angle BOC = 38^\circ$ are adjacent angles. 2

23. Apply the reverse-and-add process on 47 until a palindrome appears. 2

24. Draw (describe) a pictograph for: Guava 25, Mango 10, Banana 35 with one symbol = 5. 2

25. Test whether 3216 is divisible by 4 and by 8. 2

26. Find the area of a triangle with base 12 cm and height 9 cm. 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

27. Add the triangular numbers 45 and 55. What do you notice about the answer? 3

28. A student turns clockwise from facing North through 90°. In which direction (N, E, S, W or in between) is she now facing? Name the type of the angle of turn. 3

29. Show that the Kaprekar routine on 3524 reaches 6174. 3

INTERMIND

30. Study the table and answer: 3

(i) the largest,
(ii) the smallest,
(iii) the total.

Item	Number
Banana	40
Grapes	30
Apple	10

31. Use the divisibility test for 11 to check whether 2728 is divisible by 11. 3

OR

A rectangle is 15 cm long and 8 cm wide. Find its perimeter and area. 3

32. Find the sum of $\frac{2}{3}$ and $\frac{3}{4}$ and compare the sum with 1. 3

OR

Describe the construction of the perpendicular bisector of a line segment of length 6 cm. 3

SECTION — D (Long Answer Type) — 6 × 5 = 30

Answer in about 100–150 words each.

33. A group of 6 friends shake hands with each other exactly once. Using the idea of a complete graph, find the number of handshakes. Show the working. 5

34. A ray OA makes 30° with OB and another ray OC makes 110° with OB on the same side of OB, with OC beyond OA. Find $\angle AOC$, and name the types of $\angle AOB$, $\angle BOC$ and $\angle AOC$. 5

35. Take the digits of 3141. Form the largest and smallest numbers from them, subtract, and repeat the process twice more. Write each result and say what you notice. 5

36. The table shows data on favourite sports of students. Draw a pictograph taking one symbol = 5 students, and find the total number of symbols. Data: Tennis 15, Cricket 30, Kabaddi 35. 5

OR

Find all prime numbers between 1 and 40 using the Sieve of Eratosthenes idea. State which primes you cross out multiples of. 5

37. An L-shaped figure is made by joining a rectangle 10 cm by 6 cm and a rectangle 4 cm by 5 cm (they do not overlap). Find the total area of the figure and by how much the larger part exceeds the smaller part. 5

OR

Riya reads $\frac{1}{4}$ of a book on Monday and $\frac{1}{3}$ of it on Tuesday. What fraction has she read in two days? What fraction is left? On which day did she read more? 5

38. Construct a rectangle with sides 8 cm and 6 cm using only a ruler and compass (describe the steps). Verify by measuring that the diagonals are equal (state the property used). 5

OR

Study the shape: a regular hexagon. Write 5

(a) number of lines of symmetry,
(b) order of rotational symmetry,
(c) smallest angle of rotation,
(d) the angle after 4 such rotations.

END OF QUESTION PAPER — Total: 80 Marks